

Reading Note : Time delay deep neural network-based universal background models for speaker recognition

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ABSTRACT

This is a reading note of the paper Time Delay Deep Neural Network-based Universal Background Models for speaker recognition (D. Snyder et al., 2015). The paper explores the use of TDNN-based UBM for speaker recognition tasks. The paper compares the performance and efficiency of the TDNN-based UBM with traditional Gaussian Mixture Model (GMM)-based UBM (in the supervised and unsupervised case). My motivation for choosing this paper lies in its relevance to knowing more about the advances in speaker recognition technology, I particularly took this study because it pushed me to study traditional techniques for Speaker recognition and the incorporation of Deep Neural Networks in the same time. The insights gained from this study not only contribute to my academic pursuits but also inspired my first personal project to start with in the Embedded AI field, I found that this can be a great opportunity, as I will soon start implementing a version of TDNN-UBM model running on tiny device, and work on the overall workflow from model implementation, training, testing to integration in a MCU, to testing it in real world.

KEYWORDS

speaker recognition, deep neural networks, time delay neural networks

1 INTRODUCTION

Speaker recognition or verification involves confirming or rejecting an individual's identity base on their voice samples. Each person's voice has unique characteristics influenced by the size and the shape of their vocal cods and vocal tracts, and a lot of other factors.

This technology is primarily used for security access, transaction authentication and suspect identification, etc. Speaker verification has a rich scientific foundation, supported by over 30 years of research, development, and evaluations in biometric-based speech recognition.

Any speaker verification system deployment is composed of two main phases: training and testing, with basic schemas presented in figures 1 and 2. During the training phase, the signal is first processed through a speech detector to eliminate regions without speech, a process known as Voice Activity Detection (VAD), most of the time it's an energy-based VAD. This step is crucial because speech is inherently discontinuous, containing pauses and breaks within sentences. Removing silent parts from the voice signal can substantially lower computational costs. Once the desired voice signal is obtained, feature extraction is performed to reduce the speech signal to fewer dimensions, represented as feature vectors. This stage involves speech pre-processing and Mel-Frequency Cepstral Coefficients (MFCC), and other coefficients, but the MFCC are the most widely used for speech recognition and speaker verification. So several audio feature vectors are extracted from the speaker's speech in order to create a distinctive model that distinguishes the speaker from other speakers.

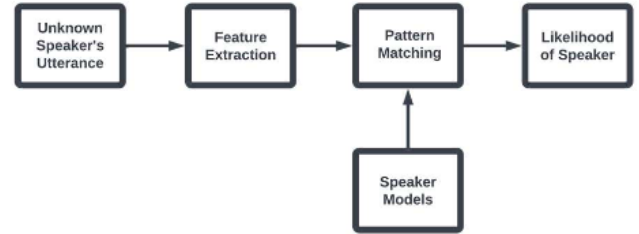


Figure 1: Basic block diagram for Automatic Speaker Verification system during the training phase.

In the testing phase, we compare the supplied speech signal to the existing model, with the help of a set of statistical models and parameters extracted during training, to compute a score that gives a decision about the identity of the speaker, i.e the voice is authenticated as the claimed speaker identity or not.

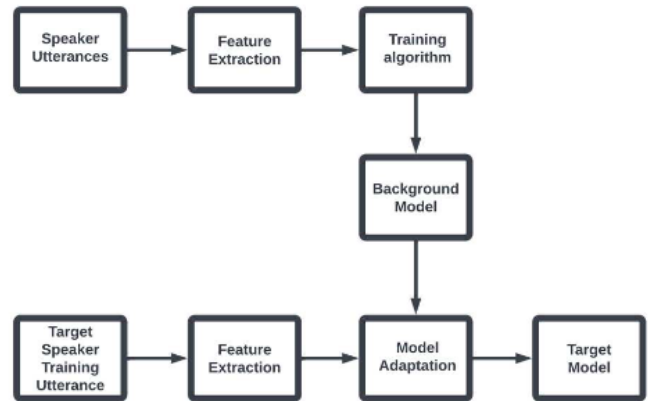


Figure 2: Basic block diagram for ASV system during testing phase

The paper in our hands experiments with the baseline GMM, supervised-GMM and Time Delay Neural Network (i.e, TDNN) based systems, to give us results on the incorporation of TDNNs in the schemas in figures 1 and 2, and explore the use of a supervised version of GMM-based Speaker Recognition system.

2 TIME DELAY NEURAL NETWORK

TDNNs [5] are widely used for acoustic models in modern speech recognition for modelling long term temporal dependencies from short-term speech features i.e., MFCCs. In a standard deep neural network (DNN), the initial layer applies a transformation to the entire temporal context at once. In contrast, a time-delay neural network (TDNN) starts by learning transformations on narrow contexts, with deeper layers gradually processing broader temporal relationships. Each layer in a TDNN operates at a different temporal resolution, which increases with depth. The transformations in TDNNs are tied across time steps, similar to convolutional neural networks. During back-propagation, this tying results in lower layers being updated by gradients accumulated over all time steps, forcing them to learn translation-invariant features.

To summarize my understanding, I can only say that *The TDNN is essentially a 1-D convolutional neural network without pooling and with dilations.*

3 GMM-UBM

This model extensively well explained in [3], has been used in speaker recognition for many years as one of the fundamental components across different frameworks. The core of the idea of a universal background model (UBM), represented by a Gaussian mixture model (GMM) trained on many different speakers, is the likelihood ratio. Say that we want to determine if speech sample X has been spoken by S . Then, the verification task is a basic hypothesis testing:

H_0 : Y is from speaker S , H_1 : Y is not from speaker S . The test to decide whether to accept H_0 or not is the Likelihood Ratio (LR):

$$LR = \frac{p(Y | H_0)}{p(Y | H_1)}$$

If the Likelihood ratio is greater than the threshold K , we accept H_0 , otherwise we accept H_1 .

If we talk in terms of logs, then the log-likelihood ratio is simply the difference between the logs of the 2 probability density functions:

$$\log(LR) = \log(p(Y | H_0)) - \log(p(Y | H_1))$$

We have a speaker to test for H_0 , and we can build a model, say λ_{hyp} , being for example a Gaussian Distribution of the features extracted.

However, we do not have an alternative model for H_1 . We must compute what is called a “Background Model”, which would be a Gaussian Model $\lambda_{hyp-bar}$. The UBM is a large GMM trained to represent the speaker-independent distribution of features. We train (EM algorithm) these models on a large dataset of speakers, each on a different speaker, before combining them to an universal background model.

4 SYSTEMS

In the paper, the comparison is done between 3 models, **baseline GMM**, **sup-GMM** and **TDNN-UBM** systems. All these systems are based on the 2 models presented in 2 and 3.

4.1 GMM-UBM Baseline

The workflow of the first system is as follows :

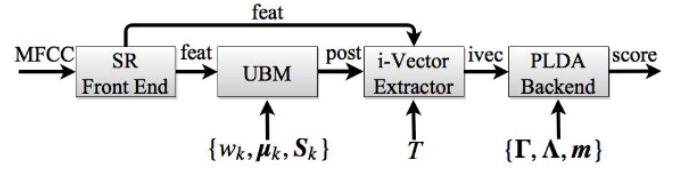


Figure 3: GMM-based speaker recognition schema.

MFCC (Mel-Frequency Cepstral Coefficients)

The process begins with the extraction of Mel-Frequency Cepstral Coefficients (MFCCs) from the audio signal.

SR Front End

- *Speaker Recognition Front End*: This module processes the MFCCs and extracts more refined features that are better suited for speaker recognition. These features are then fed into the next stage, the Universal Background Model (UBM).

UBM (Universal Background Model)

- *UBM Parameters* $\{w_k, \mu_k, S_k\}$: Each Gaussian component k in the UBM has parameters w_k (weight), μ_k (mean), and S_k (full-covariance, i.e the components may independently adopt any position and shape).

- *Function*: The UBM generates posterior probabilities for each feature vector, indicating the likelihood of the feature vector belonging to each Gaussian component in the UBM.

i-Vector Extractor

- *i-Vector Model T*: This stage involves extracting i-vectors, which are low-dimensional representations of the speaker’s characteristics. The i-Vector Extractor uses the posterior probabilities from the UBM and the extracted features to compute these i-vectors.

PLDA (Probabilistic Linear Discriminant Analysis) Backend

- *PLDA Parameters* $\{\Gamma, \Lambda, m\}$: The PLDA model is used to perform speaker verification or identification. It models the i-vectors by separating the speaker-specific and session-specific variations.

- *Function*: The PLDA Backend compares the i-vector of the test sample with stored i-vectors (from enrolled speakers) and computes a score that reflects the likelihood that the test sample belongs to a particular speaker. This score is the final output of the system.

4.2 Supervised GMM-UBM

The supervised and unsupervised GMMs differ only in the UBM training procedure, the other parts of the system follow the same baseline stated in 4.1. To train the UBM, the authors use a very different method. Using a TDNN explained in 2 and the same architecture as in the paper [2], triphone posteriors are computed on the SWB and SRE training data. An energy-based VAD removes features and posteriors corresponding to nonspeech frames. The mixture weights w_k , means μ_k , and covariances S_k are initialized according to equations 1, 2, 3 and 4. The DNN parameters are collectively labeled Θ and $P_r(k | y_i, \Theta)$ is the probability of triphone k at frame i given the DNN features y_i . The corresponding speaker recognition features are denoted x_i .

$$z(i)_k = P_r(k | y_i, \Theta), \quad (1)$$

$$w_k = \sum_{i=1}^N z(i)_k, \quad (2)$$

$$\mu_k = \frac{1}{w_k} \sum_{i=1}^N z(i)_k x_i, \quad (3)$$

$$S_k = \frac{1}{w_k} \sum_{i=1}^N z(i)_k (x_i - \mu_k)(x_i - \mu_k)^\top. \quad (4)$$

4.3 TDNN-UBM

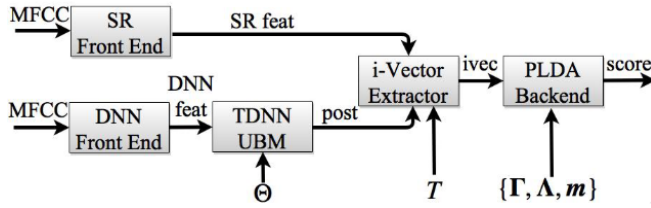


Figure 4: TDNN-based speaker recognition schema.

This system uses, like the previously mentioned, MFCCs as input and a SR front end to extract more refined features. The innovation here is that another set of parameters is extracted from the MFCCs using a DNN, in parallel with the SR front end, this parallel processing aims to capture additional information that can enhance recognition performance. Then a TDNN-UBM is trained using supervised GMM, which helps the i-vector extractor with posteriors, and the backend of the system remains the same as the first model in 4.1.

5 RESULTS

The study evaluates recognition performance at three operating points: equal error rate (EER) and normalized detection cost function (DCF) with target speaker probabilities of 10^{-2} and 10^{-3} . The EER is the point where the false acceptance rate (FAR) and false rejection rate (FRR) are equal, providing a single metric to compare different systems' overall performance. The DCF, on the other hand, is a more nuanced measure that incorporates the costs of errors and the probabilities of target speakers being present. By setting target speaker probabilities to 10^{-2} (1%) and 10^{-3} (0.1%), the DCF evaluates the system under different scenarios reflecting how often a genuine user is expected. Lower probabilities simulate more secure environments where genuine users are rare, making accurate recognition crucial. These metrics together provide a comprehensive evaluation of the system's effectiveness and reliability in various operational contexts.

Results indicate minor performance differences among unsupervised GMMs with 2048, 4096, and 5297 components, with GMM-5297 chosen as the baseline due to its slightly better EER.

Performance comparisons among GMM-5297, sup-GMM-5297, and TDNN-5297 systems show that TDNN-based systems achieve the best results, with TDNN-5297 obtaining 1.20% and 1.09% EERs for gender-independent and gender-dependent evaluations, respectively. TDNN systems show a relative improvement of 47.80%-57.59% over the baseline, while sup-GMM systems improve by 13.65%-26.55%.

The efficiency of GMM-based methods during i-vector extraction is highlighted, with the GMM-2048 system being approximately twice as fast as larger GMMs. GMM-based systems are at least ten times faster than real-time without parallelization (faster than real-time, means that it takes less time to process the audio data than the actual duration of the audio itself). The TDNN system, however, is slower due to the computational demands of feature and posterior calculations, being nearly 36% slower than real-time.

The observed results derive from the fundamental differences in the architectures and computational requirements of the GMM, sup-GMM, and TDNN systems. GMM-based systems, particularly with fewer components like GMM-2048, are computationally efficient during i-vector extraction due to their simpler modeling and lower computational demands, allowing them to operate significantly faster than real-time. In contrast, TDNN systems, while offering superior recognition performance due to their deep learning architecture, that gives them the ability to capture complex patterns in the data, require extensive computations for feature and posterior calculations, resulting in slower times. The sup-GMM system, which integrates supervised learning into the GMM framework, strikes a balance by enhancing recognition accuracy over the baseline GMM but remains less computationally intensive than the TDNN-UBM based system.

6 ADAPTED METHODOLOGY

6.1 Pros

- (1) *Comprehensive comparison*: The study compares multiple systems (GMM, sup-GMM, and TDNN) in both gender-independent and gender-dependent configurations. This comprehensive approach allows for a deep evaluation of each system's strengths and weaknesses.
- (2) *Realistic Operating Points*: Evaluating performance at realistic operating points, including equal error rate (EER) and normalized detection cost function (DCF) with target speaker probabilities of 10^{-2} and 10^{-3} , provides valuable insights into system performance in practical scenarios.
- (3) *Detailed Efficiency Analysis*: The study not only examines recognition accuracy but also delves into computational efficiency, providing real-time factors and CPU usage comparisons. This is crucial for understanding the practical deployability of these systems in real-world applications.

6.2 Cons

- (1) *Limited Dataset Partitioning*: While the study partitions data into male and female sets for gender-dependent models, it doesn't explore other demographic factors (e.g., age, accent) that might impact system performance.
- (2) *Complexity of TDNN Implementation*: The TDNN system, despite its superior performance, is computationally intensive and requires advanced hardware (e.g., CUDA for faster-than-real-time performance). This limits its practicality in resource-constrained environments.
- (3) *Lack of Generalization Testing*: The evaluation is based on the SRE10 dataset, and there is limited discussion on how well these findings generalize to other datasets or real-world applications.

7 POSSIBLE IMPROVEMENTS

The paper explores the use of Time Delay Neural Networks (TDNNs) for speaker recognition, with the TDNN-UBM system achieving significant relative improvements in accuracy but at the cost of increased computational complexity. Several potential improvements can be considered to enhance the system's efficiency and performance further.

First, all three systems (GMM, sup-GMM, and TDNN) currently run on the CPU, leading to higher computational costs and slower processing times. Leveraging GPU parallelization can significantly boost performance. TDNNs operate similarly to 1-D convolutions with shared parameters in layers, which are well-suited for parallel processing. GPUs are designed to handle parallel computations efficiently, making them ideal for accelerating operations involved in TDNN operations. By utilizing GPU parallelization, we can achieve faster processing times, making the TDNN-UBM system more practical for real-time applications and reducing the overall computational cost.

Second, another significant bottleneck in CPU execution is the handling of floating-point operations, which are computationally intensive and consume considerable memory bandwidth. Quantization involves reducing the precision of the weights and activations, which can lead to faster computation and reduced memory usage without substantial loss in accuracy. Implementing techniques such as integer quantization can convert the floating-point weights and activations into lower precision formats. This process can be carried out during the training phase, followed by fine-tuning to minimize any accuracy loss. Quantization can lead to reduced computational requirements, faster inference times, and lower power consumption, making the TDNN-UBM system more efficient and suitable for deployment in resource-constrained environments.

8 IMPACT

This paper came just after the introduction of Time delay neural networks as an architecture for efficient modeling of long temporal contexts [2], particularly these two papers [2] and [4] were submitted by the same institute, *Center for Language and Speech Processing & Human Language Technology Center of Excellence The Johns Hopkins University, Baltimore*. And the second one comes to prove the performance of TDNNs, and explore the use of them in speaker recognition tasks. These papers were of great impact, as the one introducing TDNNs has more than a 1300 citation, and a lot of these papers used TDNNs as a new architecture for speaker recognition. So this paper [4], even though it doesn't represent a breakthrough, opened the way for more innovation incorporating TDNNs with other architectures, that we can see on papers citing the paper in our hands like [1] where the authors introduce an embedded Spoken Language Understanding system based on TDNN-LSTMs.

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